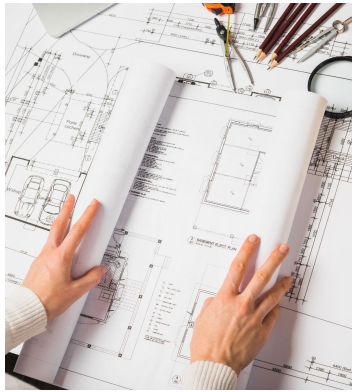
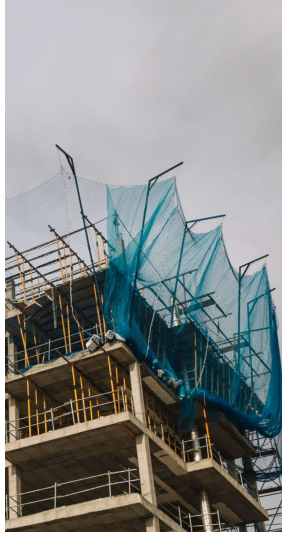
Implementing Software Architecture

Software Engineering Course – 5
2024-2025











Agenda

01

Software Architecture Recap

02

Non-functional requirements

03

Trade-off analysis

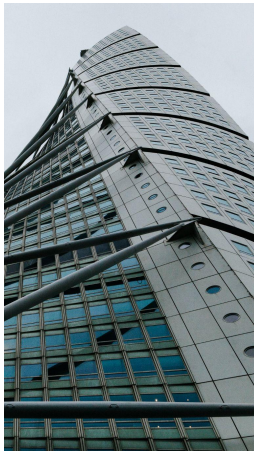
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Architecture Decision Records

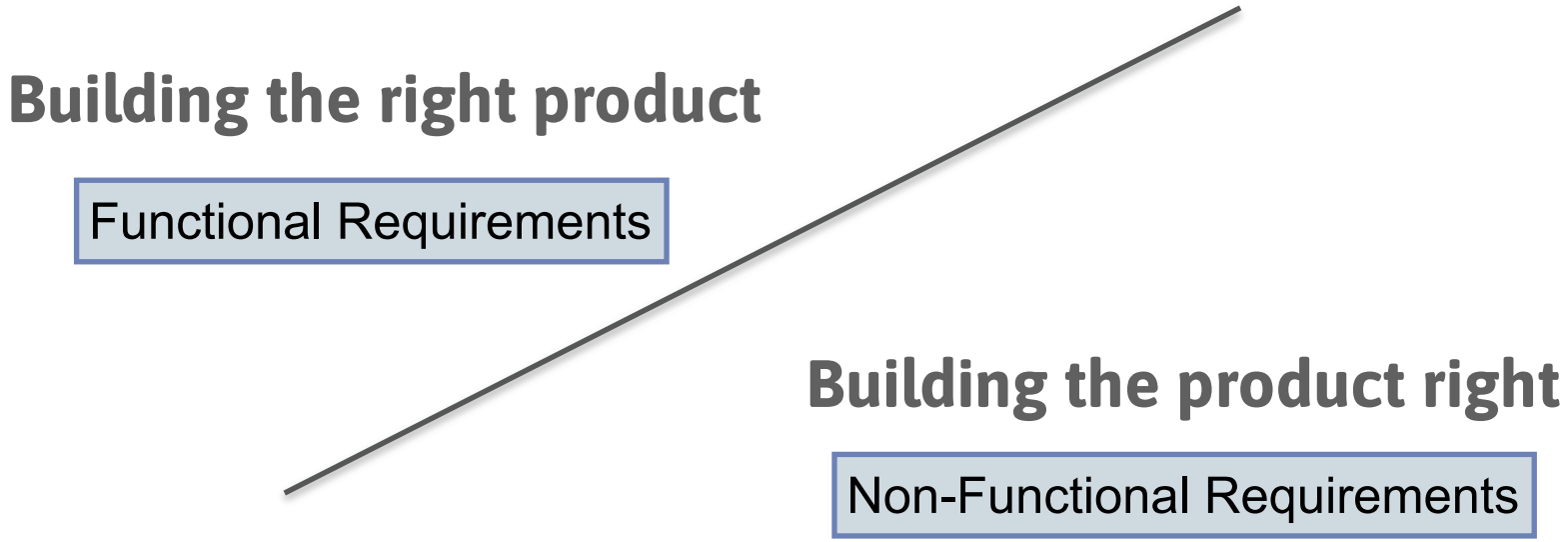
05

Architecture Pitfalls





Software Architecture

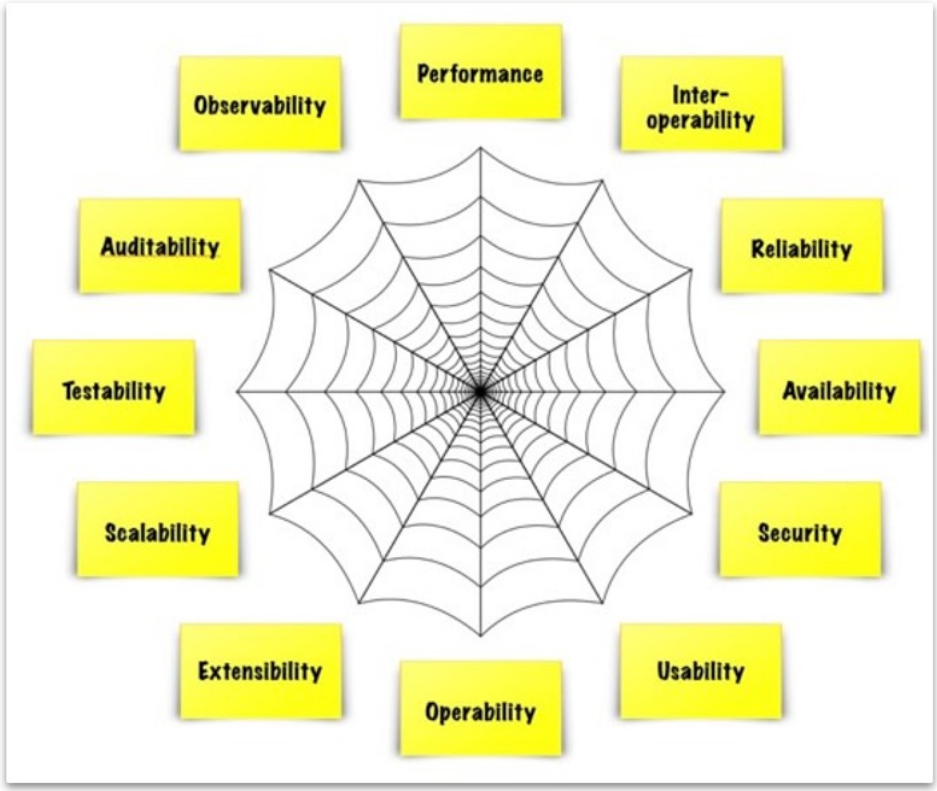


Non-Functional Requirements

Table 4.2 Non-functional system quality attributes

Attribute	Key issue
Responsiveness	Does the system return results to users in a reasonable time?
Reliability	Do the system features behave as expected by both developers and users?
Availability	Can the system deliver its services when requested by users?
Security	Does the system protect itself and users' data from unauthorized attacks and intrusions?
Usability	Can system users access the features that they need and use them quickly and without errors?
Maintainability	Can the system be readily updated and new features added without undue costs?
Resilience	Can the system continue to deliver user services in the event of partial failure or external attack?

Non-Functional Requirements



Functionality vs. Quality

Functional Requirement	Quality Attribute
New accounts should have a user defined password.	Users should immediately change the system assigned password into a user defined one Security
Create a homepage where the user can see all their information	Homepage screen should load up in less than 0.2 seconds Performance
Create a form which would process and validate user's data	The form should be modular, as the required fields are highly dynamic Modifiability

Quality Attribute Scenario

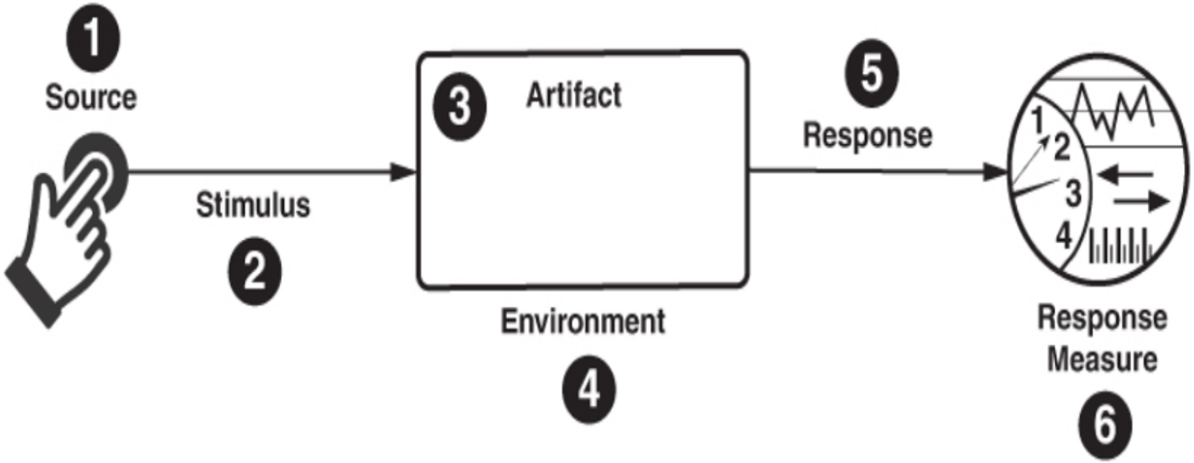


Figure 3.1 *The parts of a quality attribute scenario*

Bass, Len, Paul Clements, and Rick Kazman. *Software Architecture in Practice*. 4th edition. Boston: Addison-Wesley Professional, 2021. https://www.oreilly.com/library/view/software-architecture-in/9780136885979/.

Example: Modifiability Quality Attribute Scenario

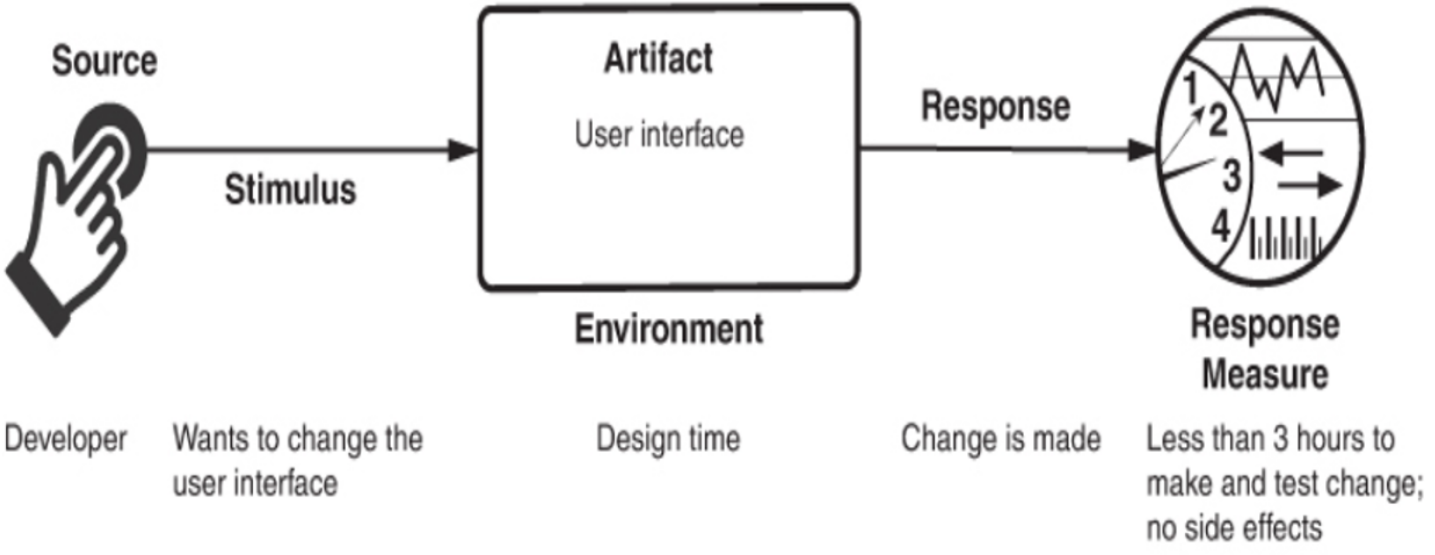
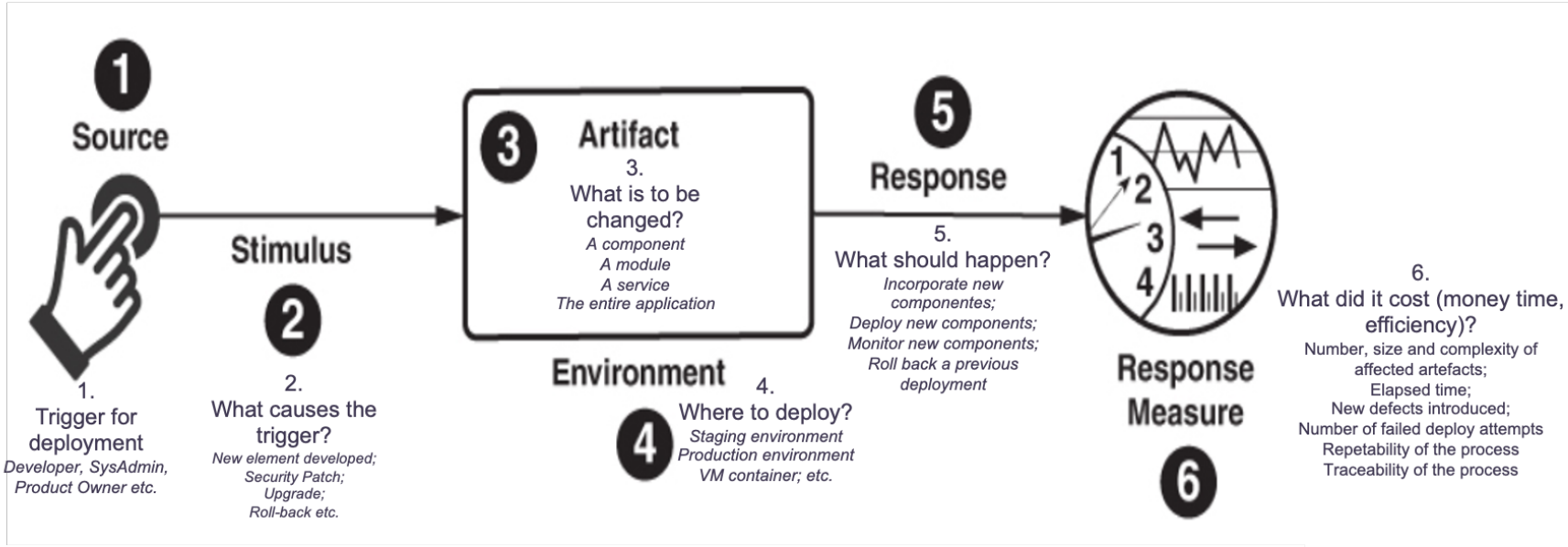
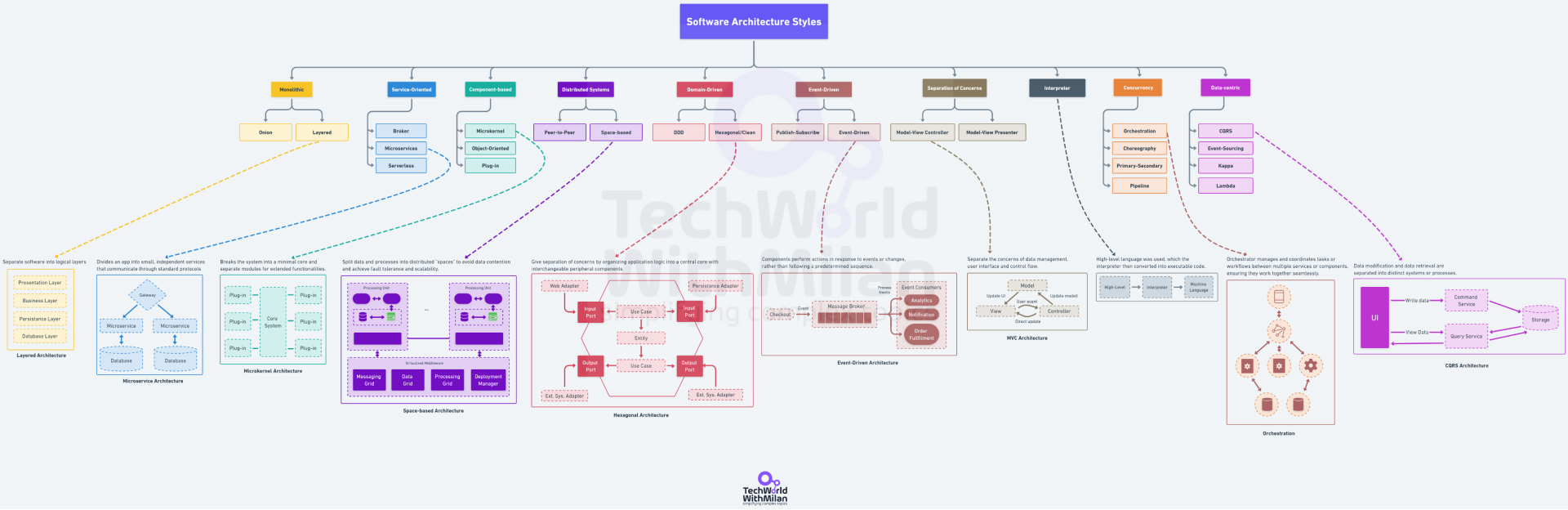


Figure 8.1 *Sample concrete modifiability scenario*

General Scenario



Software Architecture Styles



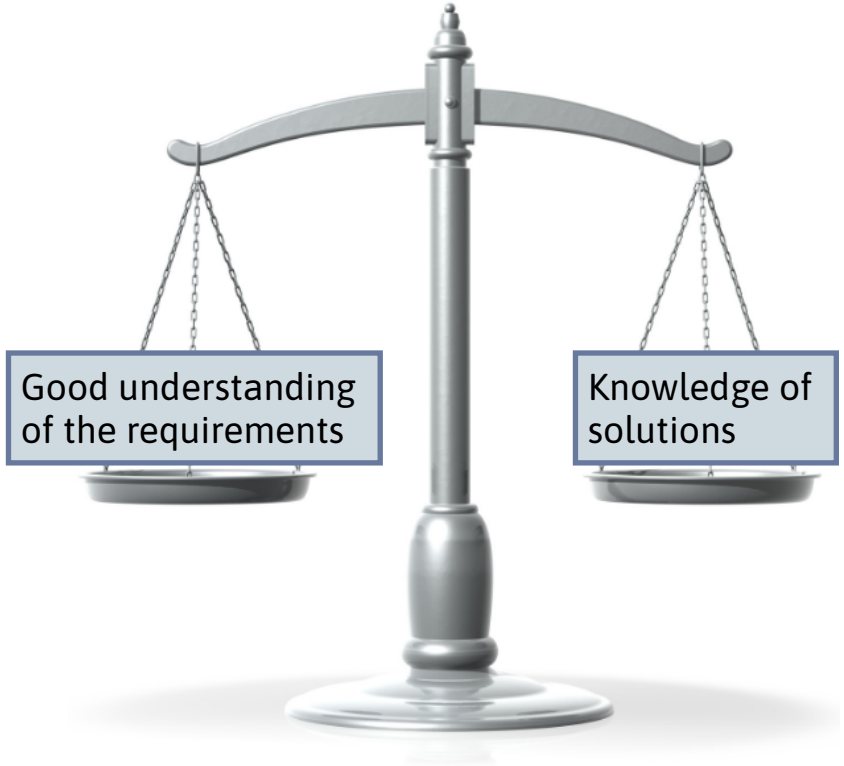
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Kahoot!

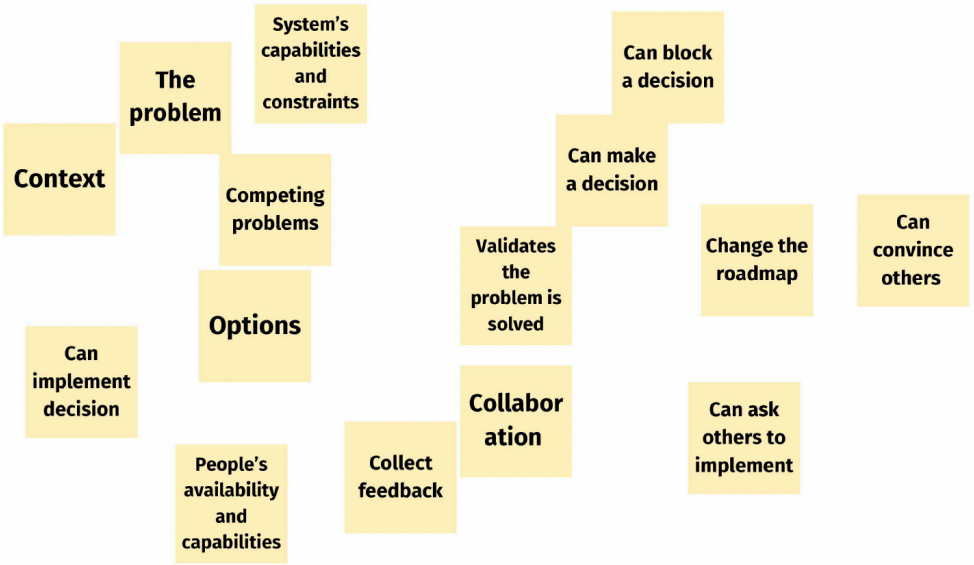
Trade-off Analysis

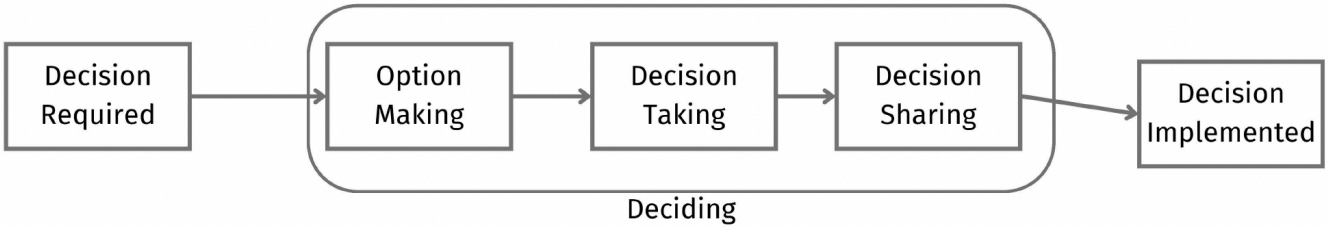


Trade-off Analysis

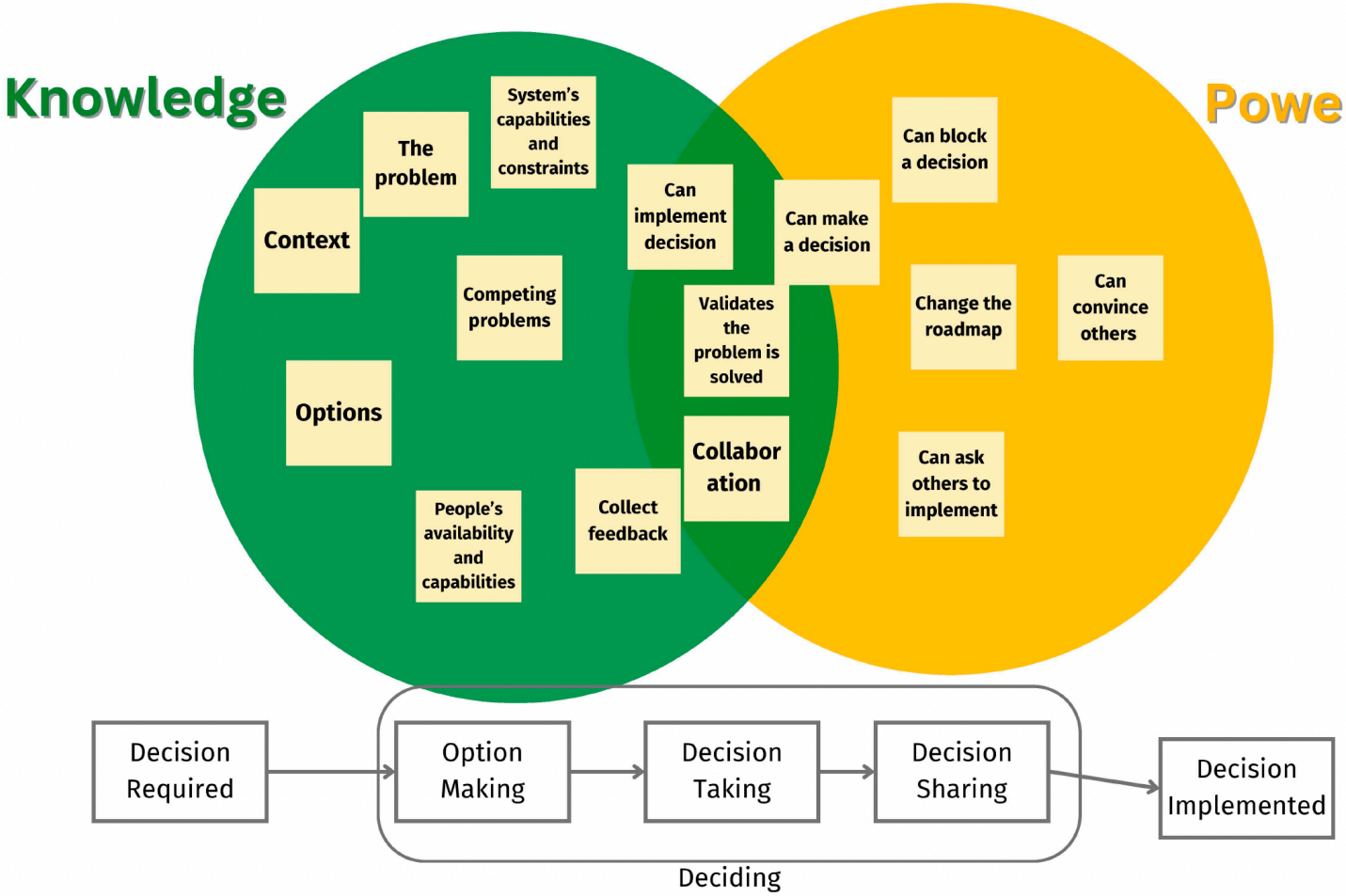


“Ingredients” of a decision

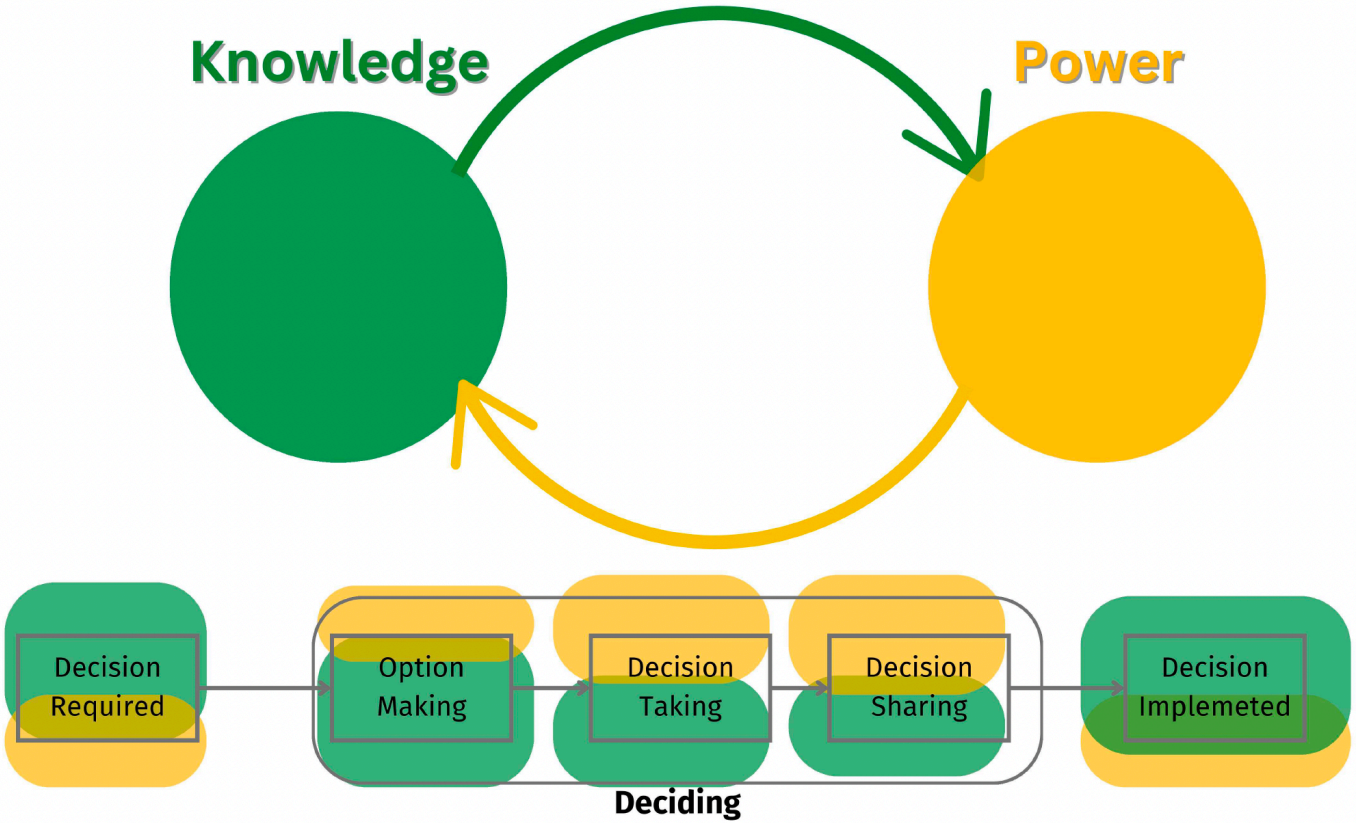




Andrea Magnorsky
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Architecture Decision Records

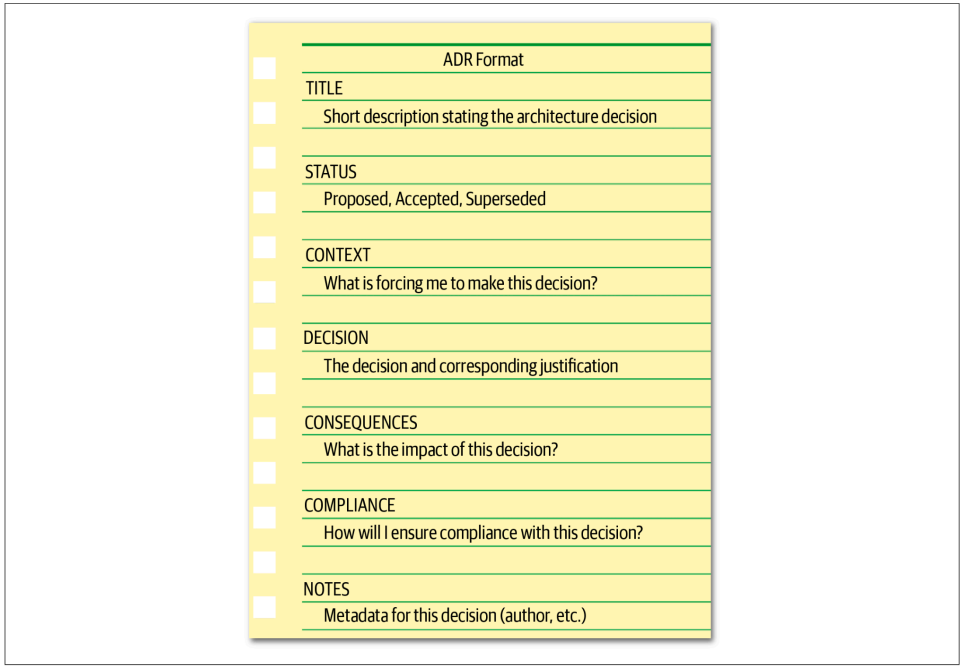


Figure 19-1. Basic ADR structure

You can find in Teams the Chapter that tells you how to build an Architecture Decision Record

Architecture Decision Records

ADR 76. Asynchronous Pub/Sub Messaging Between Bidding Services

STATUS
Accepted

CONTEXT
The Bid Capture Service, upon receiving a bid from an online bidder or from a live bidder via the auctioneer, must forward that bid onto the Bid Streamer Service and the Bidder Tracker Service. This could be done using asynchronous point-to-point (p2p) messaging, asynchronous publish-and-subscribe (pub/sub) messaging, or REST via the Online Auction API Layer.

DECISION
We will use asynchronous pub/sub messaging between the Bid Capture Service, Bid Streamer Service, and the Bidder Tracker Service.

The Bid Capture Service does not need any information back from the Bid Streamer Service or Bidder Tracker Service.

The Bid Streamer Service must receive bids in the exact order they were accepted by the Bid Capture Service. Using messaging and queues automatically guarantees the bid order for the stream.

CONSEQUENCES
We will require clustering and high availability of the message queues.

Internal bid events will be bypassing security checks done in the API layer.

UPDATE: Upon review at the April 14th, 2020 ARB meeting, the ARB decided that this was an acceptable trade-off and no additional security checks would be needed for bid events between these services.

COMPLIANCE
We will use periodic manual code and design reviews to ensure that asynchronous pub/sub messaging is being used between the Bid Capture Service, Bid Streamer Service, and the Bidder Tracker Service.

NOTES
Author: Subashini Nadella
Approved By: ARB Meeting Members, 14 APRIL 2020
Last Updated: 15 APRIL 2020 by Subashini Nadella

Figure 19-5. ADR 76. Asynchronous Pub/Sub Messaging Between Bidding Services

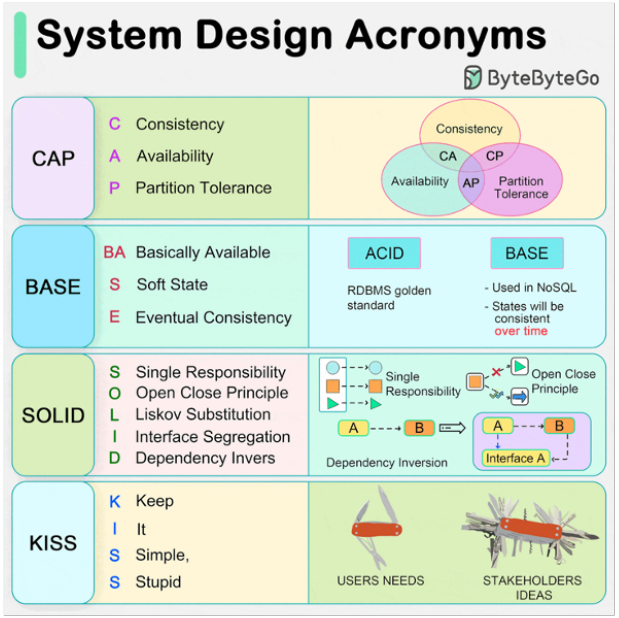
Pitfalls of Software Architecture

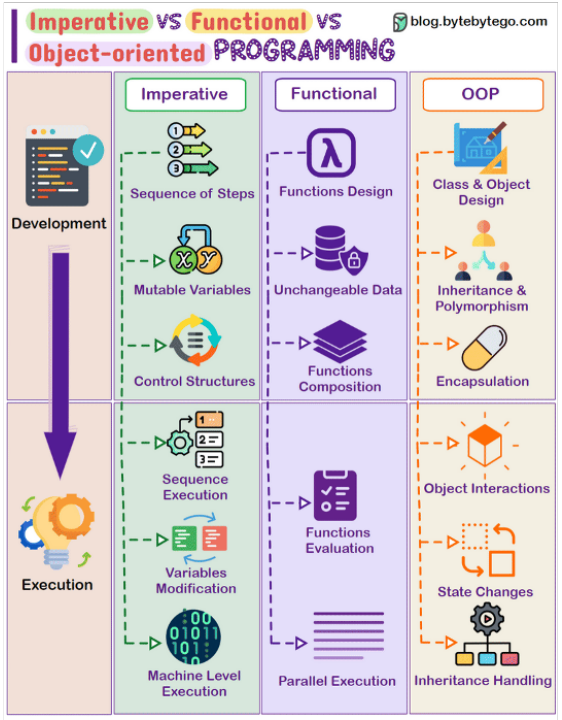
- ✓Don’t let one person make of influence all the decisions
- ✓Don’t let reuse goals dictate bad decisions
- ✓Don’t let business decisions dictate your architecture. Build NFRs.
- ✓Don’t compromise quality for faster delivery
    - ✓Create Technical Debt consciously
- ✓Don’t copy successful architecture. Design it yourself.
- ✓Don’t generalise if you aren’t sure if it needs to be generalised.
- ✓Build architecture incrementally
- ✓Publish often.

https://www.infoq.com/articles/avoid-architecture-pitfalls/

Learning resources for future Software Architects

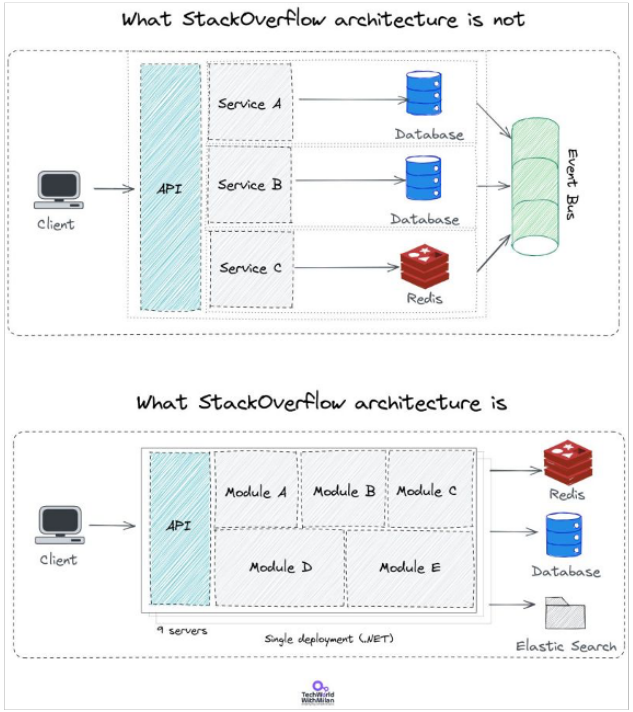
ByteByteGo
By Alex Xu

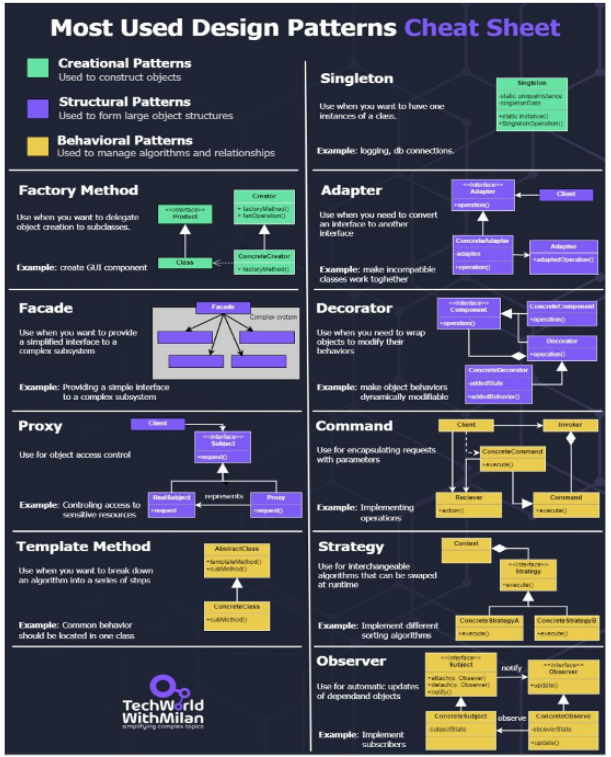




Learning resources for future Software Architects

TechWorldWithMilan
Dr Milan Milanović

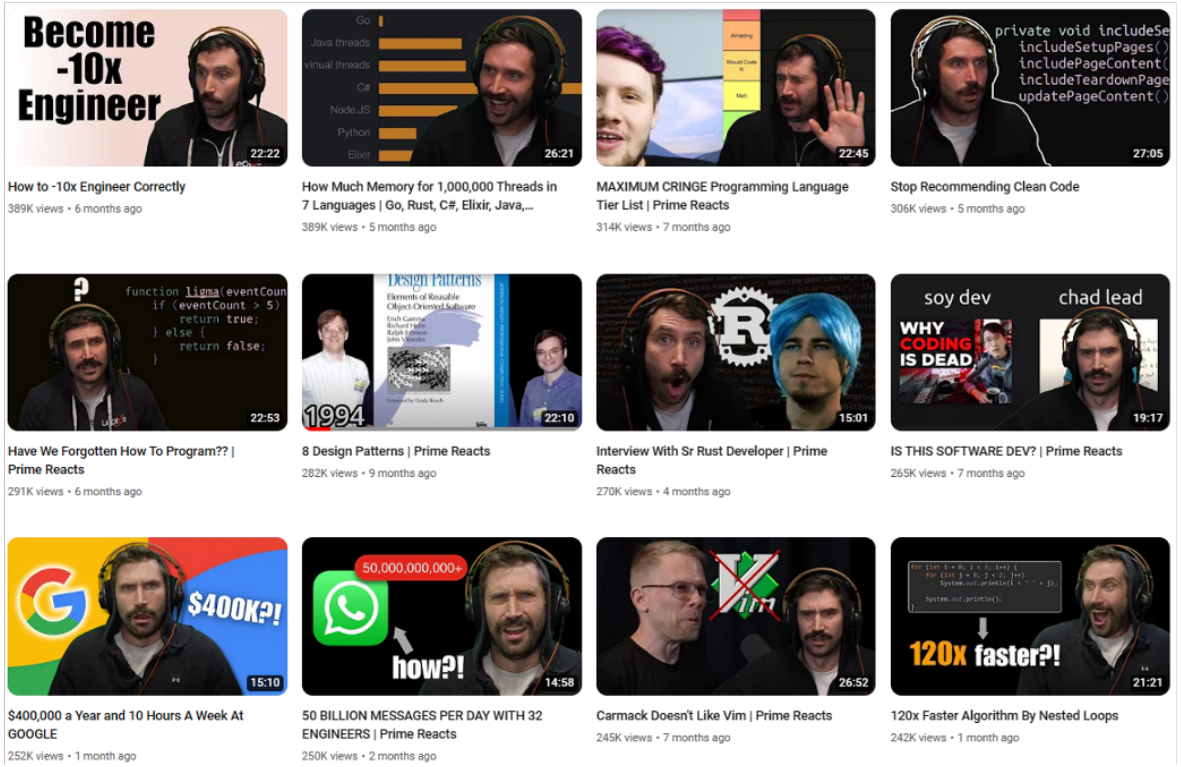




Learning resources for future Software Architects



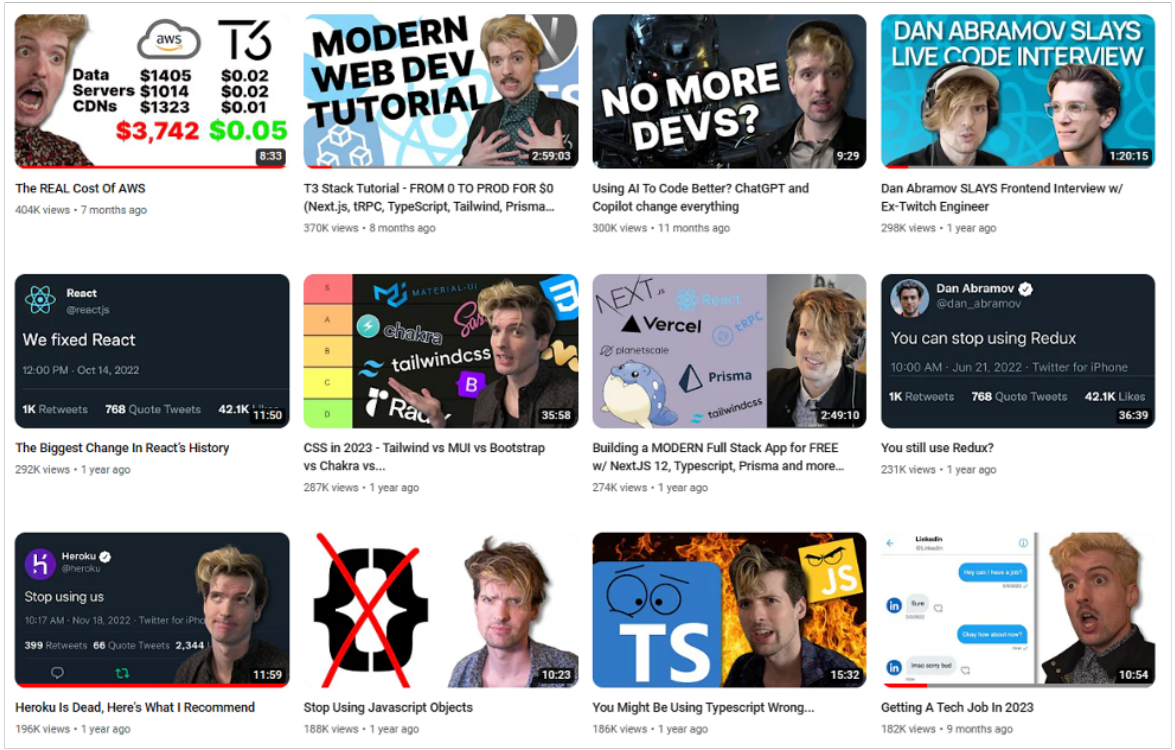
[ThePrimeTime - YouTube]



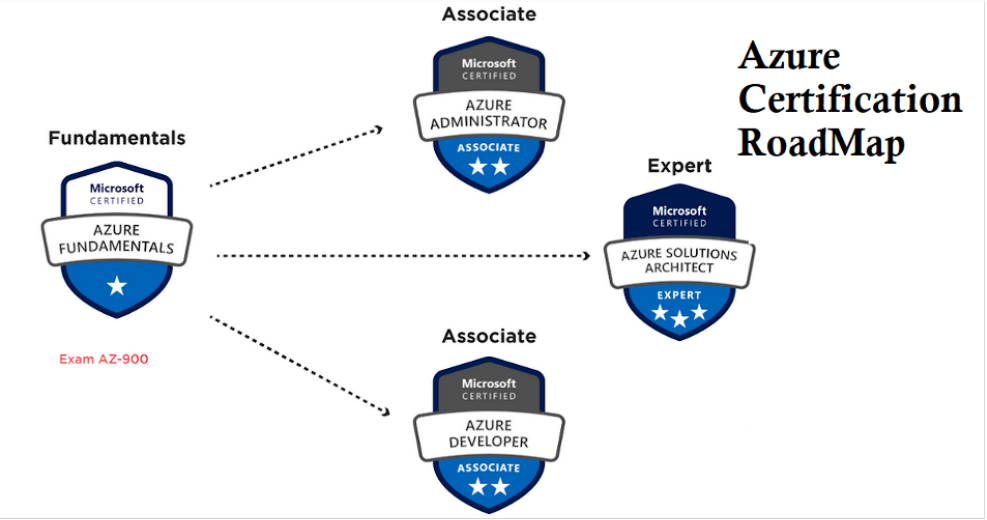
Learning resources for future Software Architects



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Courses and Certifications for Software Architects



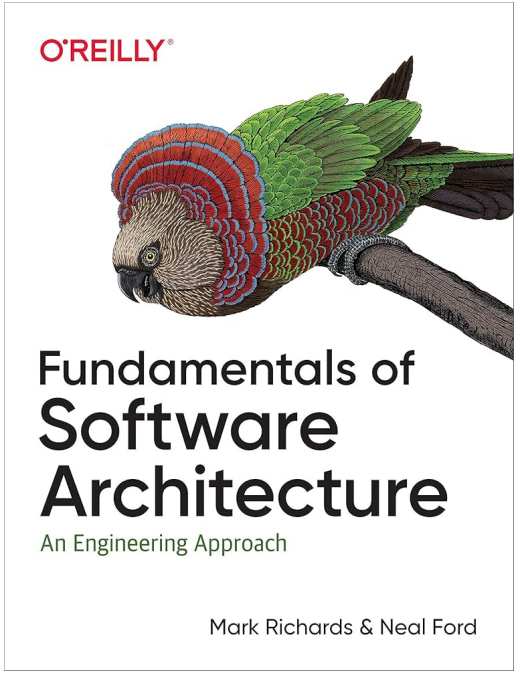


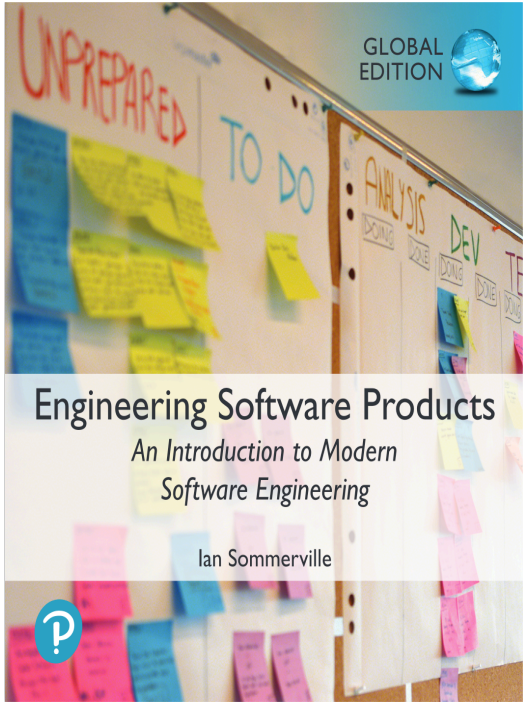
Nu vă recomand să cheltuiți banii voștri pentru astfel de certificări.
Mulți angajatori oferă în oferta de angajarea accesul la cursuri.

Books

Books aren't for learning tools/languages/syntax.
Books are for understanding patterns of thought.







Next time on …

Course 7

Please read from
Fundamentals of Software Architecutre (Mark Richards & Neal Ford):

- Chapter 19 – Architecture Decisions (separated PDF in Teams)

Engineering Software Products (Ian Sommerville)

- Chapter 4 - 4.5 Technology Issues

